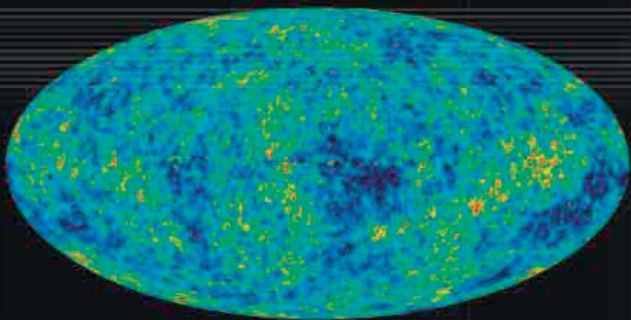




A map of microwave radiation emitted after the Big Bang defines an expanding universe that is 13.7 billion years old.

AMERICAN PHYSICISTS **Arno Penzias** (b.1933) and **Robert Wilson** (b.1936) were studying radio waves from satellites. Despite removing all known sources of interference, their antenna continued to pick up background noise. What they were hearing, by accident, was the **cosmic microwave background radiation (CMB)** left over from the formation of the universe—evidence of the Big Bang (see p.344).

For more than a century, physicists had hypothesized the

existence of astronomical objects with such compacted mass that not even light could escape from them. In 1964, these massive objects got a name: **black holes**. In June, a rocket discovered the strongest source of X-rays near Earth, **Cygnus X-1**—later shown to be a black hole. These holes are now known to be formed when massive stars collapse.

British physician **Robert Macfarlane (1907–87)**, and American scientists **Oscar Ratnoff (1916–2008)** and **Earl Davie** (b.1927) independently

“WELL BOYS, WE’VE BEEN SCOOPED.”

Robert Dicke, American physicist, on the accidental detection of microwave radiation by Arno Penzias and Robert Wilson, 1965

showed how proteins solidified in blood when exposed to air due to chemical reactions that involved different clotting factors. American physiologist **Judith Pool** (1919–75) isolated the chemical factor that was eventually used in treating hemophiliacs, people with impaired **blood clotting**.

American chemist **Jerome Horwitz** (1919–2012) made a drug called **azidothymidine (AZT)** that was a modified component of DNA. By injecting it into tumors, he hoped it would confuse cancer cells and stop them from dividing. AZT became an effective antiviral treatment for AIDS.



Alexey Leonov's historical spacewalk lasted for 12 minutes and 9 seconds.

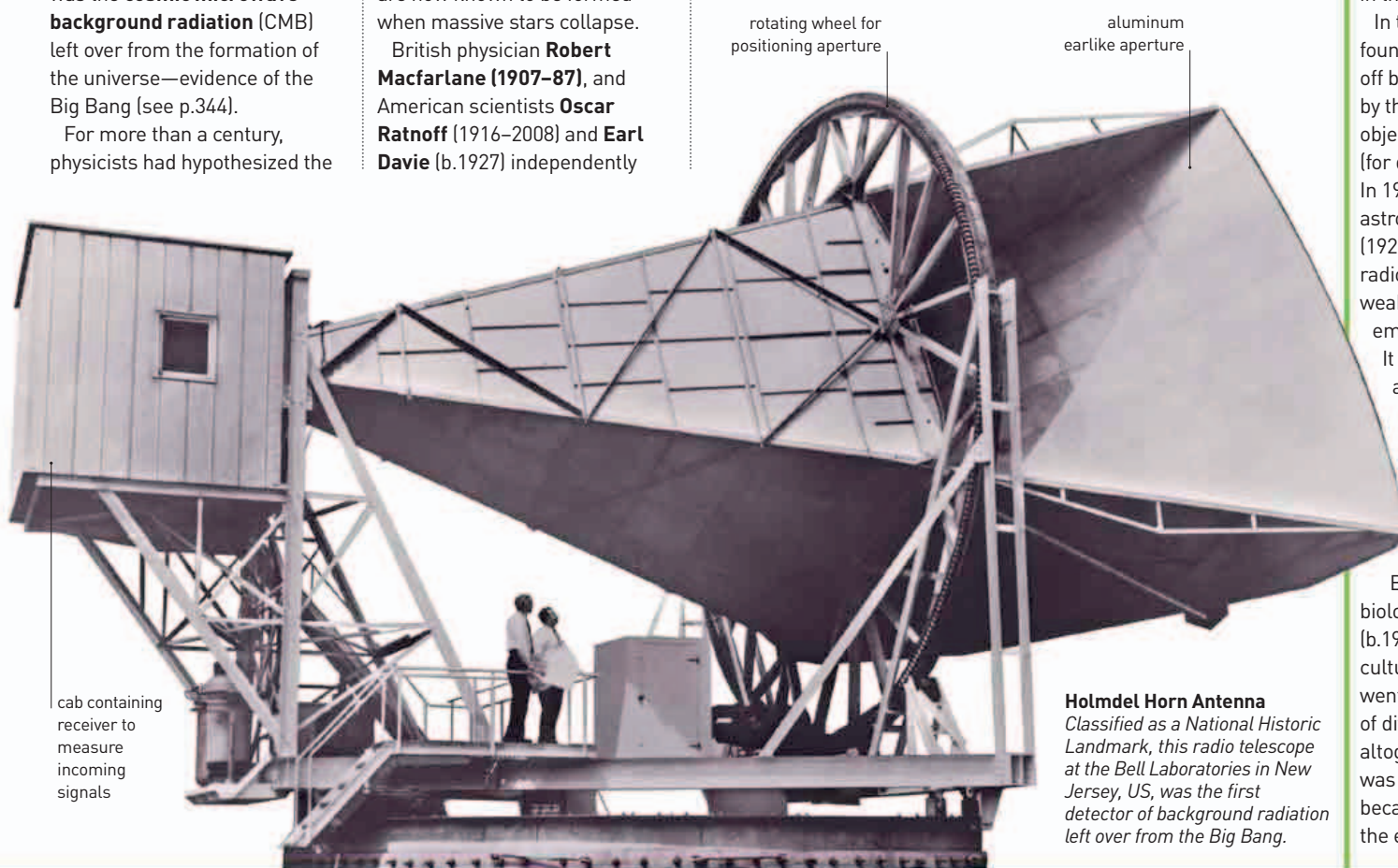
IN MARCH, SOVIET COSMONAUT ALEXEY LEONOV (b.1934) became the **first person to walk in space**. Tethered to his spacecraft, Voskhod 2, Leonov spent over 10 minutes in extravehicular activity (EVA). He nearly failed to get back inside the spacecraft because his suit had swelled up in the vacuum of space.

In the late 1950s, astronomers found a celestial object that gave off brilliant light. First detected by their radio waves, these objects were called **quasars** (for quasi-stellar radio sources). In 1965, however, American astronomer Allan Sandage (1926–2010) found the first radio-quiet quasar, which had weak radio emissions, but could emit other types of radiation.

It took another 20 years for astronomers to identify a **quasar as the core of a galaxy with a black hole at its center**.

Until 1965, biologists thought that human cells could divide continuously.

But in March, American biologist **Leonard Hayflick** (b.1928) published evidence that cultures of human cells only went through about 50 rounds of division before stopping altogether. Ten years later, it was found that this happened because each division corroded the ends of chromosomes



rotating wheel for positioning aperture

aluminum earlike aperture

cab containing receiver to measure incoming signals

Holmdel Horn Antenna
Classified as a National Historic Landmark, this radio telescope at the Bell Laboratories in New Jersey, US, was the first detector of background radiation left over from the Big Bang.

January 18 American journalist Ann Ewing first uses the term “black hole”

May Blood clotting is described in detail

June 16 US Naval Research Laboratory rocket flight discover Cygnus X-1 black hole

May American physicist Arno Penzias and Robert Wilson discover cosmic microwave background (CMB)

Azidothymidine (AZT) antiviral is developed to treat cancer and is later used for HIV treatment